Physics Note

Matsumotofukashi High School
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form 2026-04-14

1 *Electric Field*

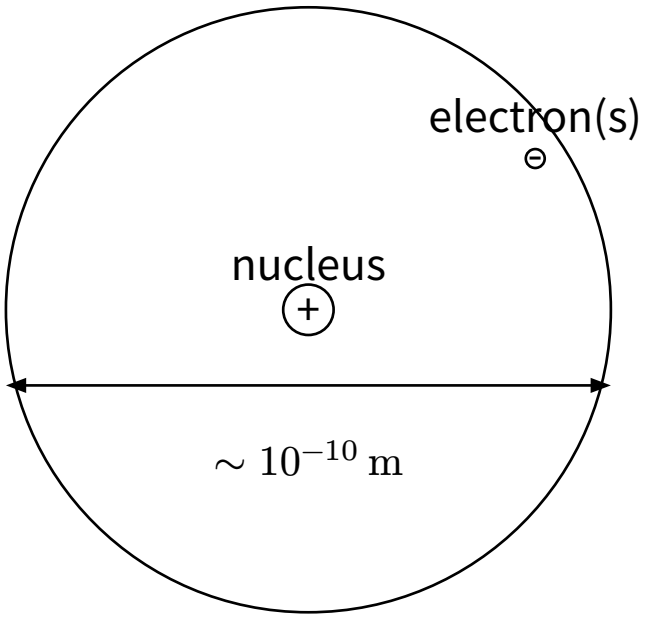
1.1 Electrostatic Force

- **electrification** - A process getting charge
- **static electoricity** - static charge
- **electoric charge** - TI
- **point charge** - charge which can be ignored the size

Electrostatic force is an interraction between charged particles.

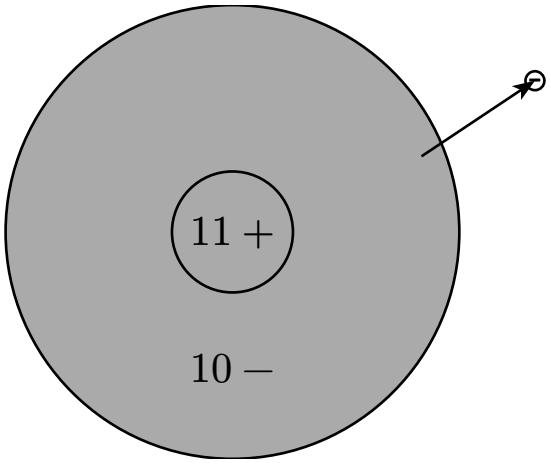
$$|\mathbb{F}| = k\frac{q_1 q_2}{r^2} \tag{1}$$

1.2 Structure of Atom



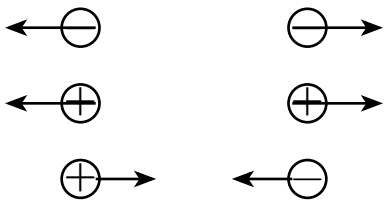
- **elementary charge** $1.6 \cdot 10^{19}\ \mathrm{C}$

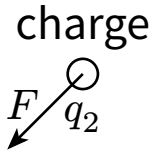
1.3 Mechanism of Electrification

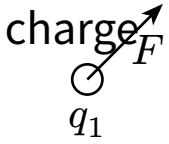


Total amount of charge is conserved.

- **Coulomb's law**







$$F = k\frac{q_1 q_2}{r^2} \tag{2}$$

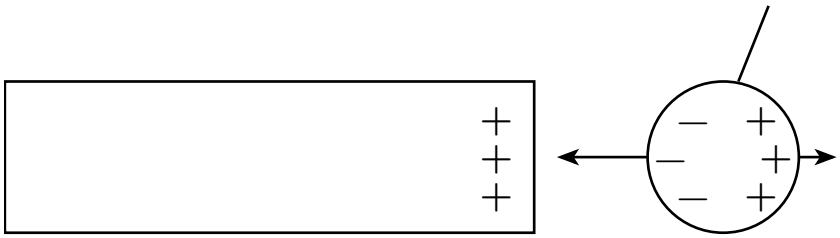
k depends on type of surrounding substance.

In bacum, $k \approx 9.0 \cdot 10^9\ \mathrm{N\,m^2\,C^{-2}}$

1.4 Electrostatic Induction

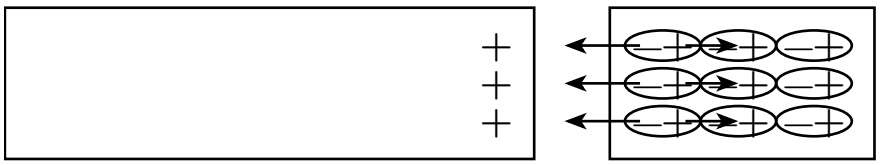
- **Conductor** - Substances which can conduct electricity.
  ex) metal, carbon
- **Nonconductor** - Substances which cannot conduct electricity.
- **Semiconductor** - Substances which have middle electric resistance between conductor and nonconductor.

- **induction**



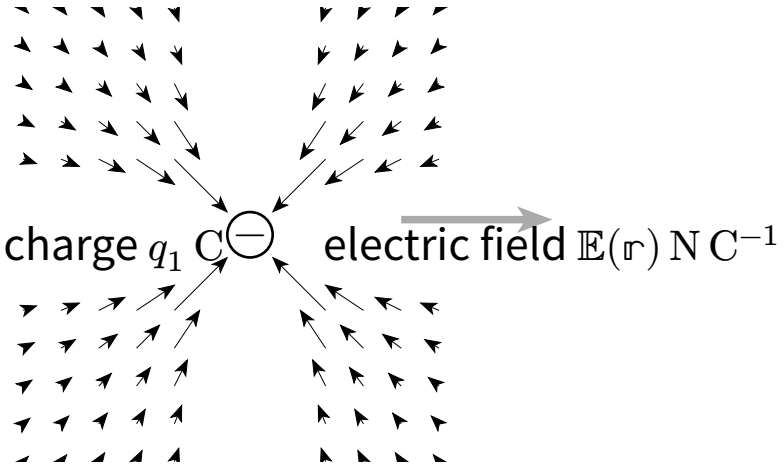
conductor

- **Dielectric Polarization**



nonconductor

1.5 Electric Field



Electric field is vector field.

Let $1\,\mathrm{C}$ charge be a test charge at position $\mathbb{r}$, $\mathbb{E}(\mathbb{r})$ is the electric force which the test charge receive.

$$\mathbb{E}(\mathbb{r}) = kQ\frac{\mathbb{r}}{|\mathbb{r}|^3} \tag{3}$$

If there are multiple charges, the electric field is equals to sum of each electric fields made by these charges.

1.6 Electric line of force

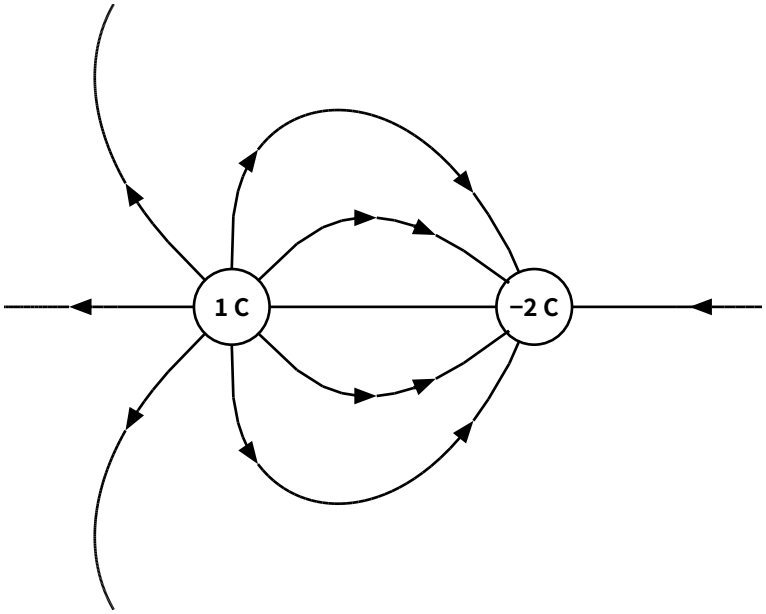
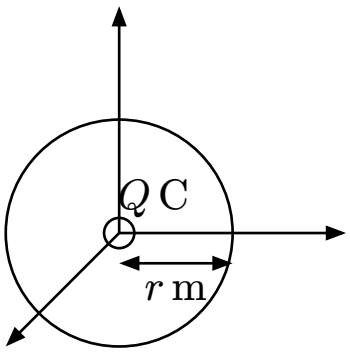


Figure 1: simulator

(it was difficult to draw the figure…)

- **the direction of tangent of line is the same as the direction of the field's vector.**
- **the line appears with plus charge and disappear with minus charge.**
- **E lines drawn per $1\ \mathrm{m^2}$, as strength of electric field is $E\ \mathrm{N\ C^{-1}}$.**



Let N be number of lines,Q amount of charge.

$N = E \cdot 4\pi r^2 = k\frac{Q}{r^2} \cdot 4\pi r^2$

Thus,

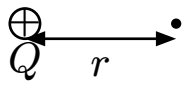
$$N = 4\pi k Q \tag{4}$$

1.7 Electric Potential

Let U potential energy of charge which have q, V electric potential,

$$V := \frac{U}{q} \tag{5}$$

Thus, V is potential energy per unit charge.



Let infinity be reference level of potential,

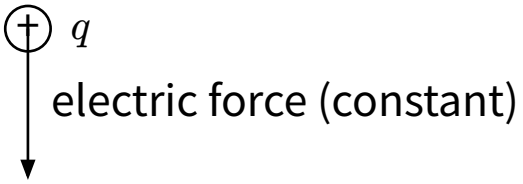
$V(r) = \int_r^\infty dr \cdot k\frac{Q}{r^2} = kQ\left[-r^{-1}\right]_r^\infty = k\frac{Q}{r}$

$$\boldsymbol{V(r) = k\frac{Q}{r}} \tag{6}$$



$V_\mathrm{AB} = V_A - V_B$ (reference is B)

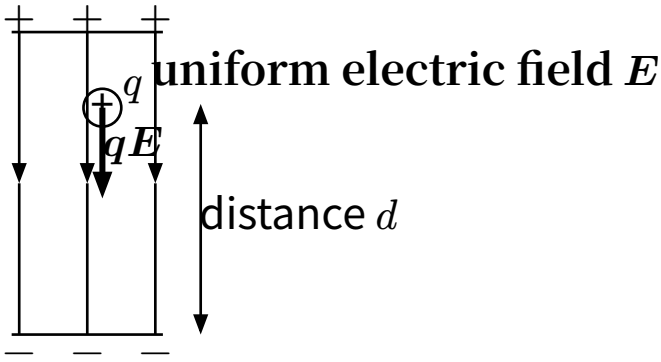
V_AB is also called $\mathbf{voltage}\left[\mathrm{ML^2T^{-3}I^{-1}}\right]$.



~~refer~~ence level of potential

$$W_\mathrm{AB} = q(V_A - V_B) \tag{7}$$

1.8 Relative of Field and Potential



$$\begin{aligned} qEd &= qV \\ \Leftrightarrow V &= Ed \end{aligned} \tag{8}$$

electric potential is proportional to energy, so it's superposition is calculated by sum.

1.9 Equipotential Surface

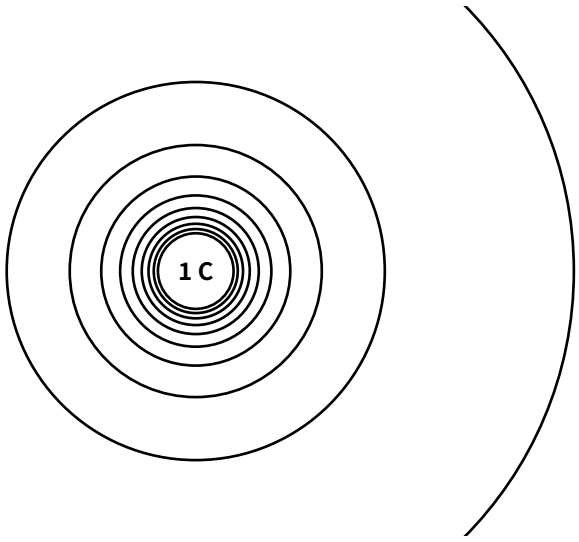


Figure 2: simulator

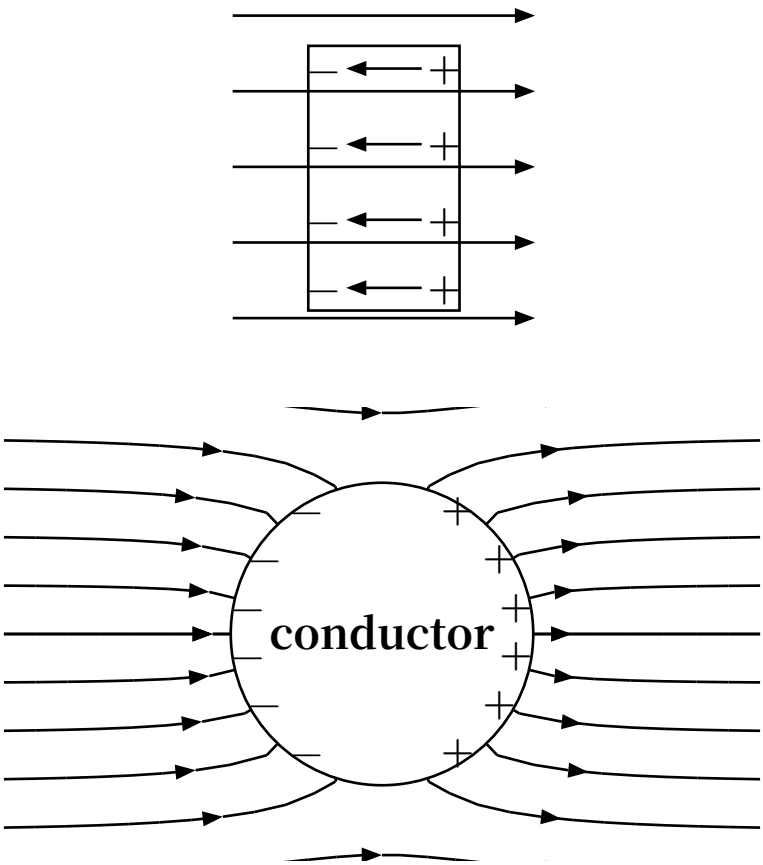
- **Equiipotential Surface** - A surface which was made by joyning points that have equal electric potential.

equipotential surface is perpendicular to electric line of force.
(if test charge move with a direction perpendicular to electric force , work is $W = \mathbb{F} \cdot \Delta\mathbb{x} = 0 \Leftrightarrow$ potential is same constantly.)

$$\frac{1}{2}mv^2 + qV = \text{const.} \tag{9}$$

1.10 Substance and Electric Field

1.10.a Conductor



In conductor, the electric field is 0. (charges contiune moving while field is not 0,)

It means that electric potential is the same in metal.

- **Electric field is perpendicular to surface of metal.** (metal distorts surrounding electric field)
- **Electric charges do not appear in internal of metal.** (on surface only)

(why? is not impossible to make field 0 with it having charges in metal? そうかな…そうかも…引力と斥力があるから自動的に表面にでできるかね。べつに全電荷が打ち消しに貢献する必要はなく、正負が surface に交互に並んでいればいい？)

- **Electric Shielding** - In cavities in conductor, electric field is ~ 0. It means field in the cavity is not affected by the outside field.

(これマジでなんで ??)

1.10.b Nonconductor

In nonconductor, the electric field is smaller thant the ouside field. (due to the electric field generated by dielectric polarization)

1.11 Capacitor

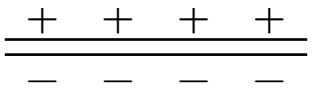


Figure 3: parallel-plate capacitor

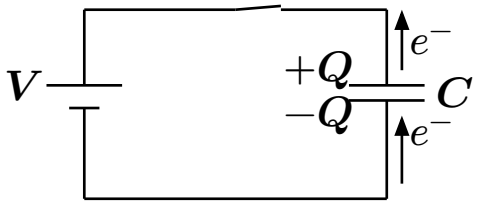


Figure 4: charging

$$C := \frac{Q}{V} \tag{10}$$

C is called **electric capacity** $[\mathrm{M}^{-1}\mathrm{L}^{-2}\mathrm{T}^{-2}\mathrm{I}^{2}]$

なにこに次元は…電磁気の次元解析ってけっこう魔境…？独自文字で置いてやろうか

Parallel-plate capacitor makes uniform electric field.

It makes $4\pi kQ$ lines of electric force .

(電気力線って superposition できるの？uniform であることは積分で証明？)

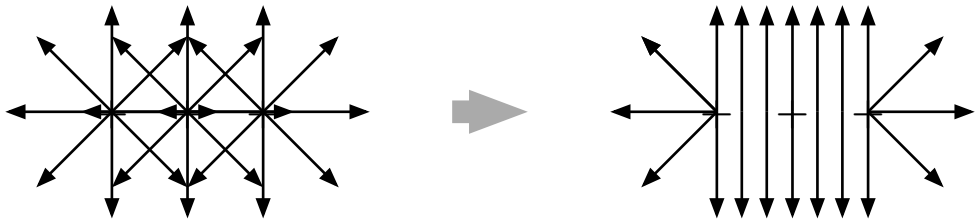


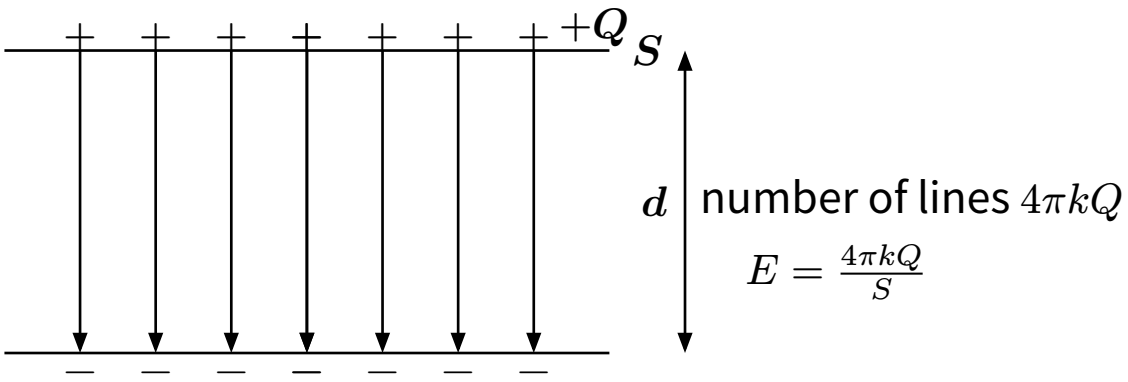
Figure 5: superposition of electric field which made by lined charges

Let

surface area of one side plate $\boldsymbol{S}$,

amount of electric charge $\boldsymbol{Q}$,

distance between plates $\boldsymbol{d}$



$V = Ed = \frac{4\pi kQ}{S}d$

$Q = \frac{1}{4\pi k} \cdot \frac{S}{d} \cdot V$

$$\therefore \boldsymbol{C = \frac{1}{4\pi k} \cdot \frac{S}{d}} \tag{11}$$

Electric capacity depends on k , S ,and d.

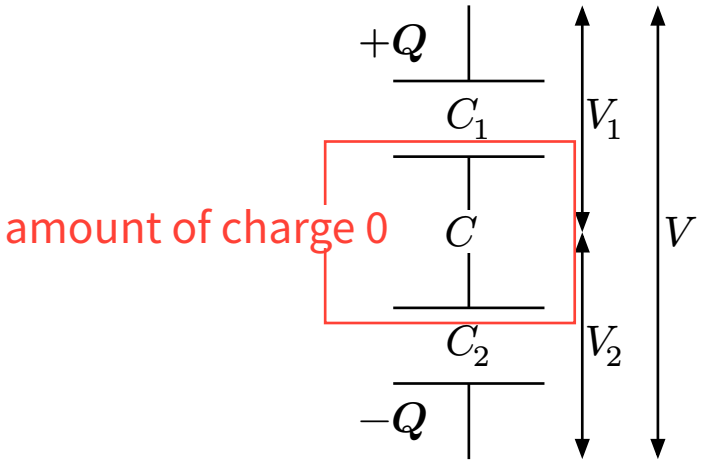
Where, let $\varepsilon = \frac{1}{4\pi k}$, ε is called permitivity $[\mathrm{M}^{-1}\mathrm{L}^{-3}\mathrm{T}^{4}\mathrm{I}^{2}]$.

$$C = \varepsilon \cdot \frac{S}{d} \tag{12}$$

In a lower permittivity substance, strength of the electric field is smaller. From $V = k \cdot \frac{Q}{r}$, even with the same voltage, the amount of charge is greater.

dielectric constant : $\varepsilon_r = \frac{\varepsilon}{\varepsilon_0}$

1.11.a Series Connection



The amount of charge is 0, because one side plate of each is connected by the conductor.

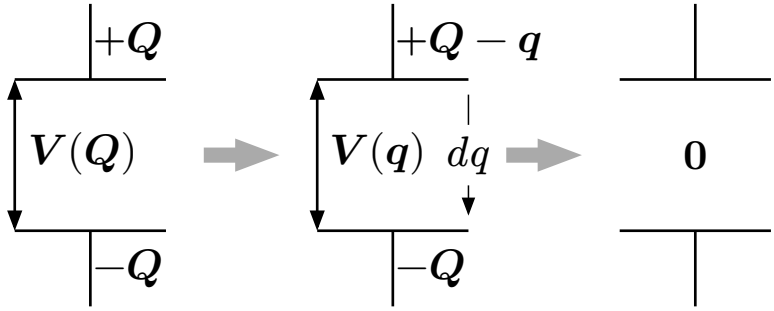
$V = V_1 + V_2$

$\frac{Q}{C} = \frac{Q}{C_1} + \frac{Q}{C_2} \left(\because V = \frac{Q}{C}\right)$

$$\therefore \boldsymbol{C^{-1} = C_1^{-1} + C_2^{-1}} \tag{13}$$

- V is proportional to C^{-1}

1.11.b Energy of capacitor



$$V(q) = \frac{q}{C}$$

$$W = \int_0^Q dq \cdot V(q) = \int_0^Q dq \cdot \frac{q}{C} = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}QV \tag{14}$$

$$\therefore \boldsymbol{W = \frac{1}{2}QV} \tag{15}$$

The work which battery has done :$W = QV$. (**V=const.**) Therefoe Work:$\frac{1}{2}QV$ is wasted.

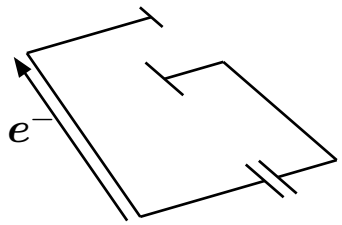


Figure 6: The start of charging

2 *Electric Current*

2.1 Current

Electric Current[I] - A flow of charged particles.

- **Direct Current** - a current that is one-directional
- **Alternating Current** - a current that periodically reverses direction

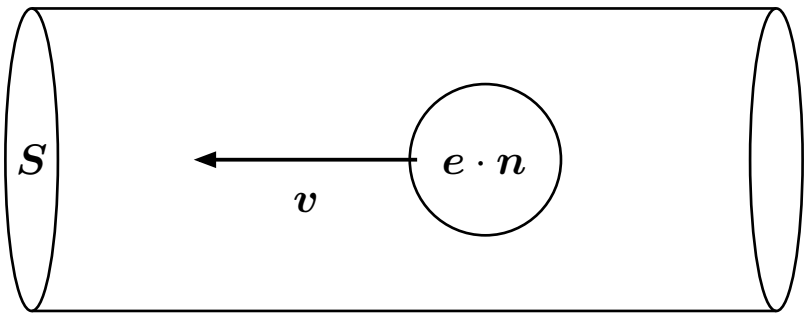
I : current (constant), Q : amount of charge, t : time

$$I := \frac{Q}{t} \tag{16}$$

$\mathrm{A} = \mathrm{C\,s^{-1}}$

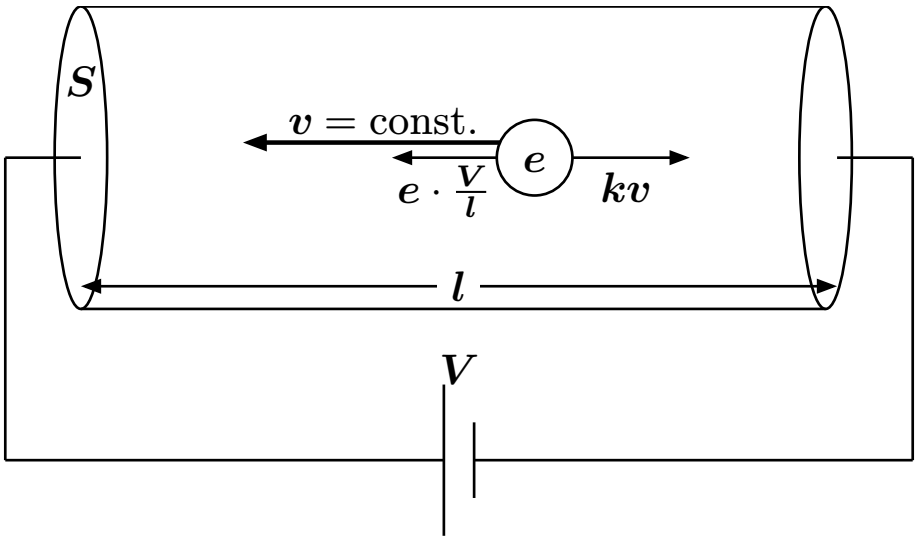
2.2 Ohm's law

Let elementary charge e , speed of charges v , number of charges per unit volume n , surface of wire S ,



$Q = e \cdot n \cdot vt \cdot S$

$\therefore I = envS$



There shoud be force $kv(k = \text{const.})$ which against electric force $e \cdot \frac{V}{l}$, because velocity of electron $v = \text{const.}$

※ energy が低い≠安定

$$kv = e\frac{V}{l}$$

$$\Leftrightarrow v = e\frac{V}{lk} \quad (v \propto E)$$

$$\Leftrightarrow I = en \cdot e\frac{V}{lk} \cdot S \quad (\because I = envS)$$

$$\Leftrightarrow I = e^2 n\frac{S}{kl} \cdot V$$

Set $k\frac{l}{e^2 nS}$ to R,

$$I = \frac{V}{R}$$

$$\therefore \boldsymbol{V = RI} \tag{17}$$

V : a.k.a voltage dump

$R = \frac{1}{e^2 n} \cdot \frac{l}{S}$ Let $\frac{1}{e^2 n}$ as ρ,

$$\boldsymbol{R = \rho\frac{l}{S}} \tag{18}$$

ρ is called **resistivity** and depends on substances. (resistivity of alminium is $\sim 2.7 \cdot 10^{-8}$)

ちうがくせいのころよく分からなかったのが解消されていってたのしー !!

いまさらだが保存力ってけっこう難しい．field を解析してどうなったら保存力なんだろう．位置にのみ依存することが必要条件になってはいそう．十分条件はなんなの　直感的な理解が足りていません。百足らず様がお通りになられたせいでしょうか。

Resistivity also depends on templature due to thermal motion.

This is related to ratio of electron's moving speed to poisitive ion's moiton speed.

constant な電圧をかけたら，抵抗値は時間発展するの？する？

$$\boldsymbol{\rho(t) \approx \rho_0 \cdot (1 + at)} \tag{19}$$

ρ_0 is resistivity at $t = 0$, t is templature. α is called **templature coefficient of resistivity**.

たまに千石電商の服きてたらついに指摘されてちょっとうれしい. 普通のこの服はデザインセンスがあると思う. 高いだけある

2.3 Joule heat

W : generating heat by current - **Joule heat**

$$\boldsymbol{W = QV = IVt} \tag{20}$$

- **Electrical Energy**$[\mathrm{MLT}^{-2}]$(電力量) - Work which is done by current(charge)
- **Electrical Power**$[\mathrm{MLT}^{-3}]$(電力) - work per unit time

教員：電力量という訳はうれしくない。日本語って science に向かないがち。　←ワイトもたまにそう思います

$$\boldsymbol{P(\mathrm{power}) = IV = I^2R} \tag{21}$$

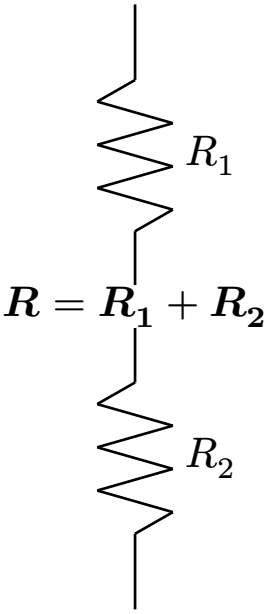
work which done by field $W = e \cdot \frac{V}{l} \cdot vt \cdot nSl = envS \cdot V \cdot t = IVt$

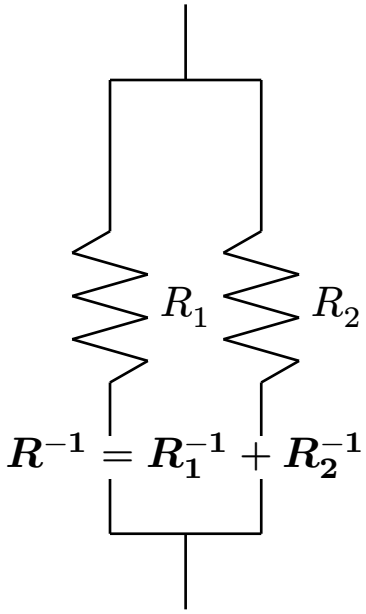
- Thermal energy は、定義が“ambiguous”なため議論には使えない(使いづらい)

heat は「macroscopic な力学で説明できない energy transfer」と定義されている

“heat が発生する”という記述ができる　高温な物体は何かしらの方法(emision など)で“heat を発生させる”から、“抵抗のある導体に電流が流れると**熱が発生する**。これをジュール熱と呼ぶ。”という記述ができるっぽい。「熱」って高校物理で直感的じゃない定義ナンバーワンじゃないか…？

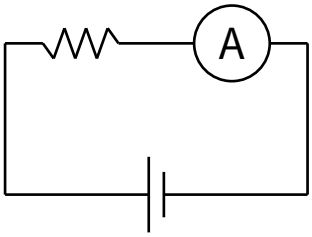
2.4 Composite of Resistors

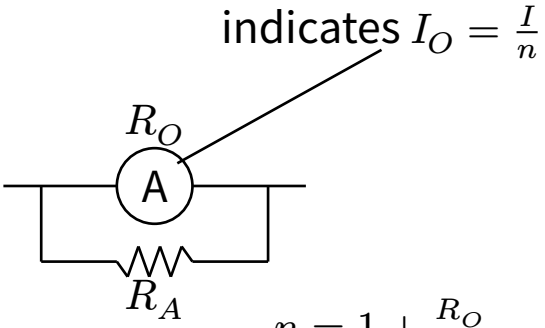




2.5 Ammeter and Voltimeter

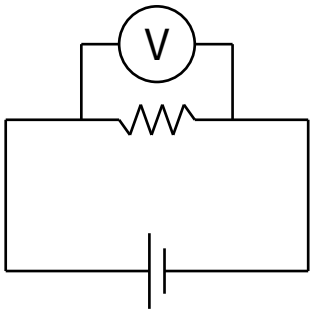
2.5.a Ammeter

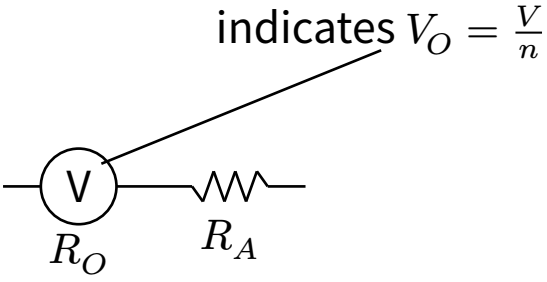




$$n = 1 + \frac{R_O}{R_A}$$

2.5.b Voltimeter

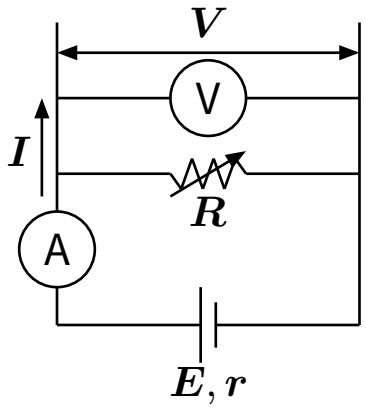




$$n = 1 + \frac{R_A}{R_O}$$

2.5.c How to Determine Electromotive Force

- **Electromotive Force(EMF) - “起電力”**

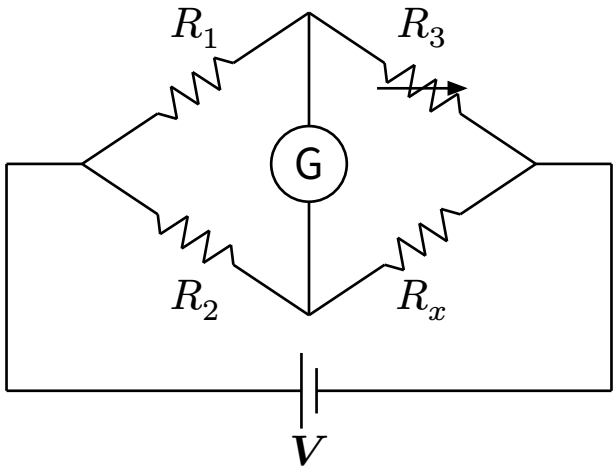


$V = E - rI$

Observing multiple (V, I) points, we can determine E and I.

2.5.d How to Determine Resistance

Wheatstone bridge



When current at G equals to 0,
$R_1 I_1 = R_2 I_2$

$R_3 I_1 = R_x I_2$

$\Leftrightarrow$

2.6 Charge of Capacitor

When a capacitor is connetced with a battery, charge of capacitor is proportional to $\exp(-\frac{t}{RC})$ (R is resistance of circuit, C is capacity)

2.7 Semiconductor

- **Intrinsic Semiconductor** 真性半導体 - Simple substance of **Si** or **Ge**. In low templature, they cannot conduct electron. In high templature, They can conduct electron.
- **Extrinsic Semiconductor** 不純物半導体 - Substance which have trace amount of **P** or **Al** in **Si** or **Ge**.

- **n-type semiconductor** (negative)

P have $4+1$ electrons on outermost shell.

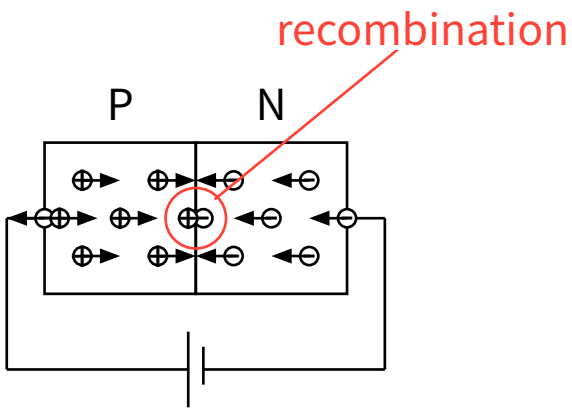
$\Rightarrow$ They have extra electrons.

- **p-type semiconductor** (positive)

Al have $4-1$ electrons on outermost shell.

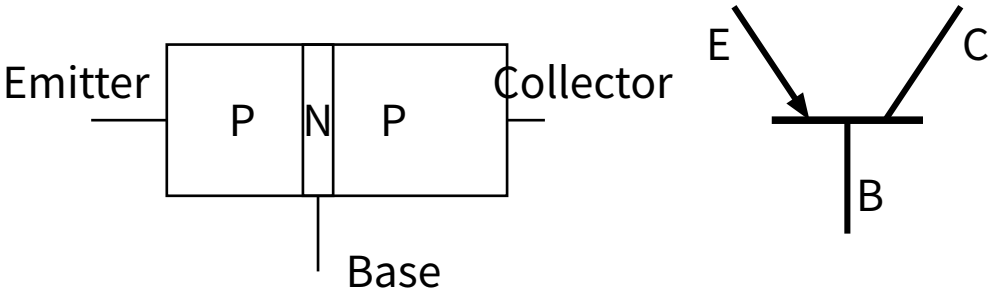
$\Rightarrow$ They have hole of electrons (positive hole). (なんでこれ共有結合できるの？の ?? オクテッドソクみたさないじょん)

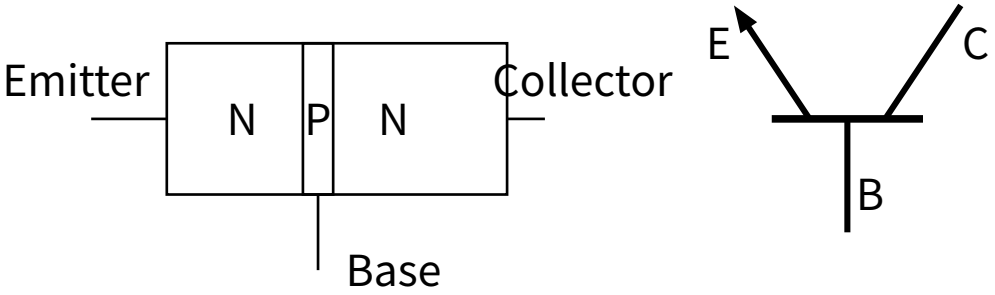
2.7.a P-N Junction / Diode



2.8 Transistor

2.8.a P-N-P Transistor





- Base semiconductor is very thin.

(なんでこれスイッチングできる ?? It's too dificult to understand.)

(P-N Junction すると，recombination（再結合）がある程度起きて平衡になる．そのとき N→P への電場が 発生していて，そのおかげで Emitter から Collector には電子が移動できない．

その電場を Emitter-Base 間の電圧で相殺することで，ポテンシャルの壁が消えて流れるようになる．

ってこと？Emitte-Base 間に電流が流れてしまうのは副産物的なもの？

なお Base 層が薄いのは，potential だけじゃ説明できなくて，electron/hole が失われる量を

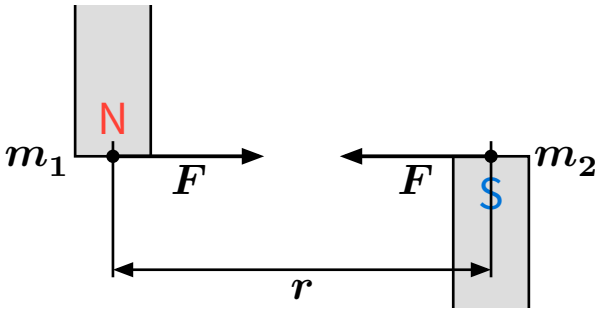
防ぐため，と理解(あってるのだろうか…))

(ついでに電池というものを「酸化還元の動的平衡」と捉えるべきであると思った 電流が流れ続けるのを LeChatelier の原理で説明できるし，電圧が発生するのも非常に直感的．教科書を書き換えよう))

3 *Electric Current and Magnetic Field*

3.1 Magnetic Field

- **magnetic pole** - **N**(north pole) and **S**(south pole) exist in pairs. This pair called magntic pole.
- **magnetic charge** [Wb] - the strength of ability to generate magnetic field. **north pole have positive value** and south pole have negative value.



$$F = k_m \cdot \frac{m_1 m_2}{r^2} \tag{22}$$

m_1,m_2 is **magnetic charge**.

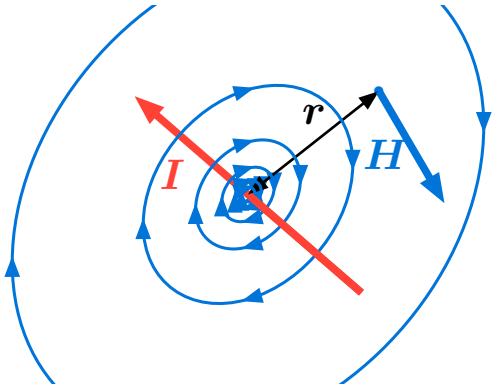
(磁力 $= mH$この式はかなり微妙らしいので書かなかった. どちらかというと磁力 $= mB$らしい. そもそも磁荷という存在が嘘みらしい)

3.1.a Magnetization

- **magnetization** 磁化 - the phenomen of being polarized into N ans S by magnetic field.

- **ferromagnet** 強磁性体 - substances which are magnetized strongly.
- **paramagnet** 常磁性体- substances which are magnetized slightly.
- **diamagnet** 弱磁性体- substances which are magnetized in the reverse direction.

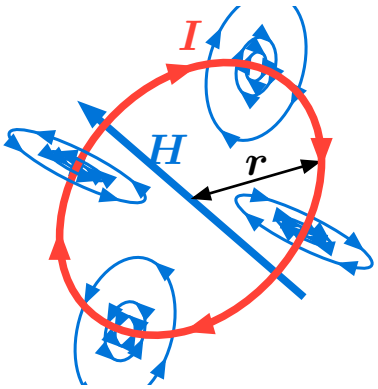
3.1.b Magnetic Field generated by Electric Current



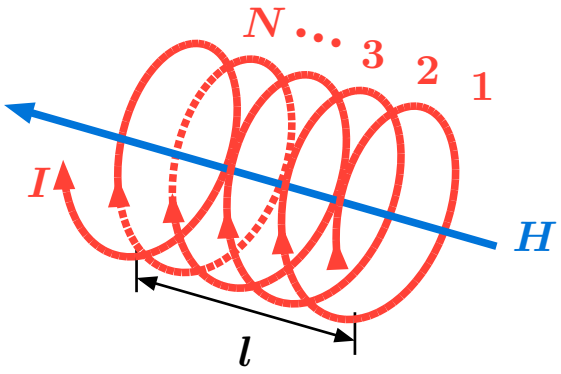
Let $\boldsymbol{H}$ magnetic field, $\boldsymbol{r}$ distance from wire.

$$\boldsymbol{H} = \frac{\boldsymbol{I}}{\boldsymbol{2\pi r}} \tag{23}$$

Thus, $[\mathrm{N/Wb}] = [\mathrm{A/m}]$



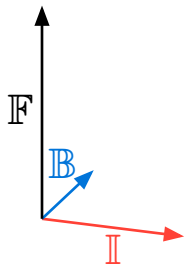
$$\boldsymbol{H} = \frac{\boldsymbol{I}}{\boldsymbol{2r}} \tag{24}$$



This is called **solenoid**.

$$\boldsymbol{H} = \frac{\boldsymbol{N}}{\boldsymbol{l}}\boldsymbol{I} \tag{25}$$

3.1.c Force generated by Electric and Magnetic field



$$\mathbb{F} = \mathbb{I} \times \mu\mathbb{H} \cdot l \tag{26}$$

$\mathbb{F}$ is force. The force depends on constant μ.

μ is **permeability**, which depends on substances.

In vacuum, $\mu = 1.26 \cdot 10^{-6}\,\mathrm{N\,A^{-2}}$

In ferromagnet, spin of electrons become aligned and amplify magnetic field $\rightarrow \sim \mu\mathbb{H}$. つまり実際に磁場は強くなっている.

In a uniform substance space, Let $\mathbb{B} := \mu\mathbb{H}$,

$$\mathbb{F} = \mathbb{I} \times \mathbb{B}l \tag{27}$$

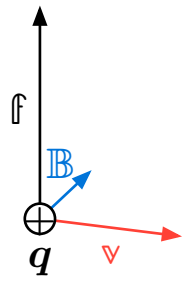
$\mathbb{B}$ is called **magnetic flux density**.

(unit : Tesla $[\mathrm{T}] = [\mathrm{N} \ / \ (\mathrm{A\ m})] = [\mathrm{Wb/m^2}]$)

(**magnetic flux**[Wb] is $|\mathbb{B}|S$.)

単位でやや混乱…[N] = [N A^-2] [A] [A m^-1] [m] μかけたら磁場が磁束密度になるんだ，磁力線の密度を磁場にしたせいきあ．[N/Wb]=[A/m]が直感的ではなさすぎる 何これー！何何これ！(ほんとになんだよ)

3.1.d Lorentz Force



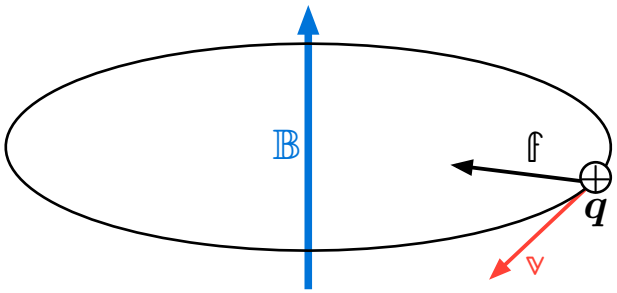
When a charged particle(q) is moving in magnetic field $\mathbb{B}$
$\mathbb{F} = \mathbb{I} \times \mathbb{B}l$

$= \mathbb{v} \times \mathbb{B}qnsl \quad (\because \mathbb{I} = qns\mathbb{v})$

There are nSl particles in the wire. (n, S, l is definited as same as Section 2.2).

Thus,
$\mathbb{f} = (qnS\mathbb{v}) \times \mathbb{B} \cdot \frac{l}{nSl}$

$= q \cdot \mathbb{v} \times \mathbb{B}$

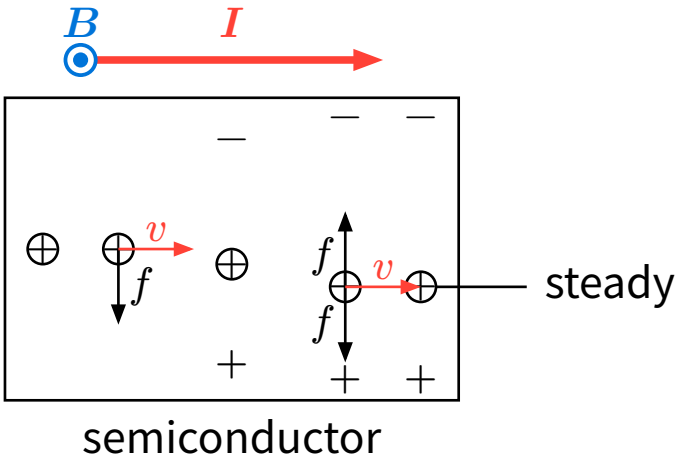


In uniform magnetic field $\mathbb{B}$, charged particles show a circlar motion.

$\omega = \frac{q}{m}B \quad \Leftrightarrow \quad T = \frac{2\pi m}{qB}$

$R = \frac{v}{B}\frac{m}{q}$

3.1.e Hall effect

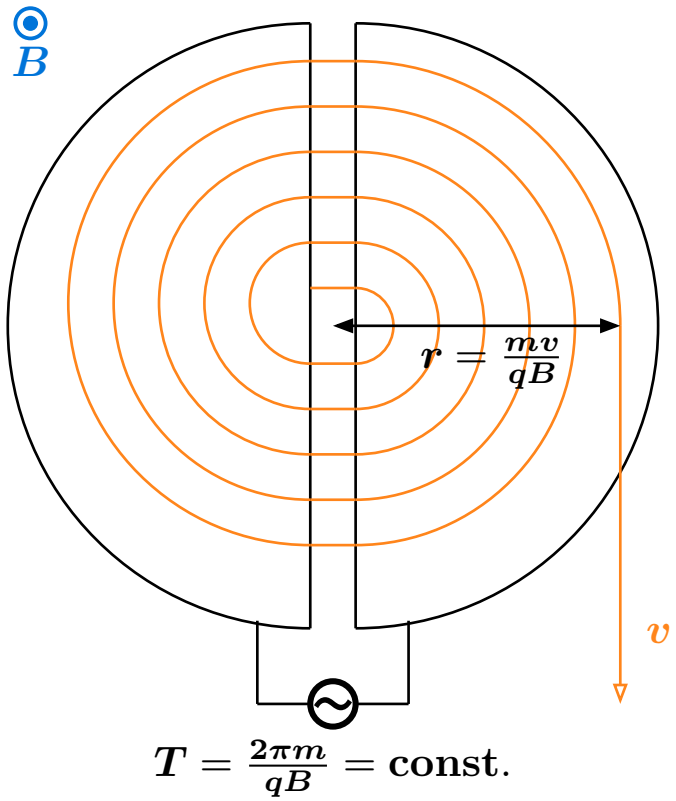


semiconductor

In uniform magnetic field, when carriers of semiconductor move, the particles experience Lorentz force and move, creating electric field.

Eventually, they get steady state. The voltage in this state called **Hall Voltage**. Which face becomes a high voltage depends on the sign of the carriers.

3.1.f Cyclotron



cyclotron is a type of accelarator.

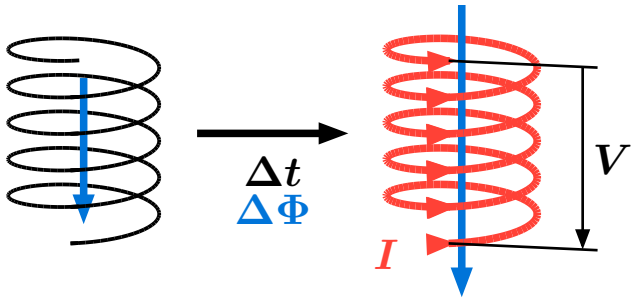
The particle receive energy from electric field which created by A.C. power source and accelarated . This system is simple because period of A.C. $T = \text{const.}$

There are also $r = \text{const.}$ accelarator which is called **synchrotron.**

(ナノテラスってすごいらしい．普通に歩いて回れる半径だったけどすごいのか)

4 *Electromagnetic Induction and Electromagnetic Wave*

4.1 Electromagnetic Induction



Faraday's law of induction (rule of thumb)

$$V = -N \cdot \frac{\Delta\Phi}{\Delta t} \tag{28}$$

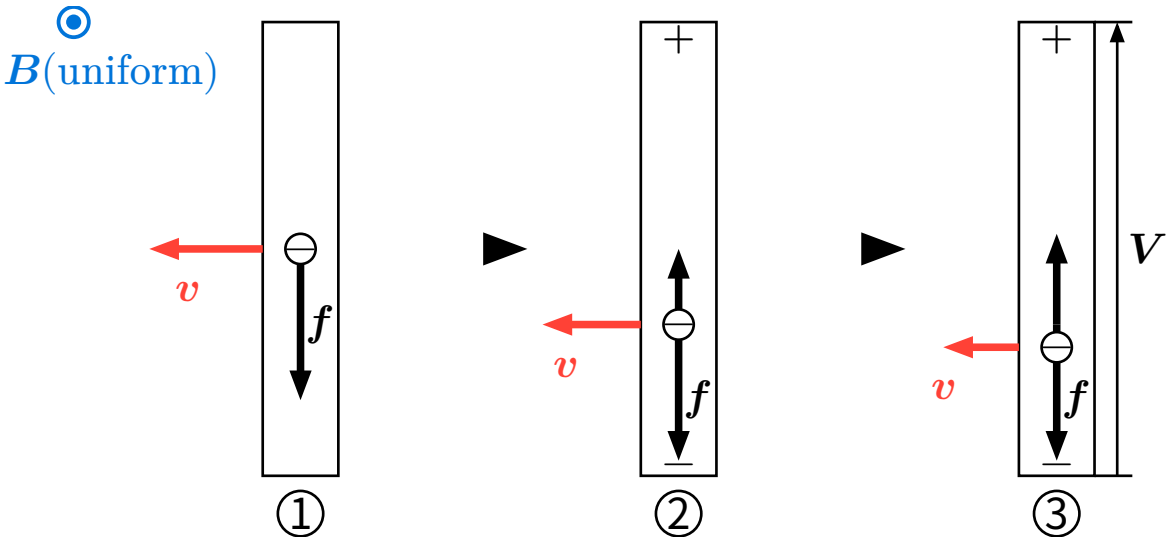
V is called induced electromotive force(**induced EMF**)

N is number of roll of coil(solenoid).

direction is such that it cancels out $\Delta\Phi$

($\Phi = S \cdot B$) (n is number of turns a the coil)

Be careful about sign. The coil work as battery, so **electrons gain potential energy** from $\frac{d\Phi}{dt}$

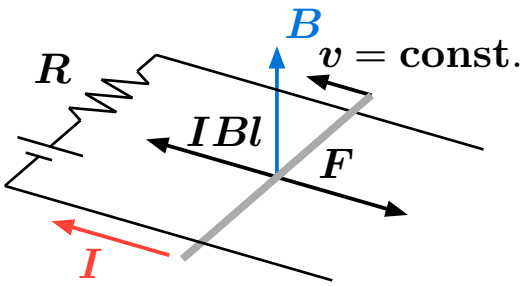


When wire move(v) in uniform magnetic field B, electrons in wire receive Lorentz force $f = evB$. This vertical force changes the direction of v. Then, and Voltage(V) generated. (3)

(In this steady state, coulomb force = lorentz force, thus, $e\frac{V}{l} = evB$)

$$V = vBl \tag{29}$$

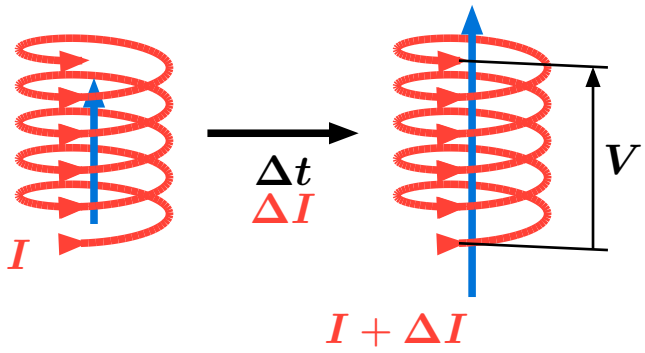
(We can obtain this from Faraday's law of induction)



$IV_0t = vtIBl + I^2Rt$

(わざわざ描くものでもなかったが練習になったのでよし)

4.1.a Self-induction of coil



$\Delta\Phi = \mu\frac{N}{l}S\Delta I \quad (\because$ Equation 25 $)$

$V = -N \cdot \frac{\Delta\Phi}{\Delta t}$

$\quad = -N \cdot \frac{\mu\frac{N}{l}S\Delta I}{\Delta t}$

$\therefore V = -\frac{\mu N^2 S}{l} \cdot \frac{\Delta I}{\Delta t}$

Let $L := -\frac{\mu N^2 S}{l}$,

(まあまあ複雑な constant を当然のように置くから困る)

$$\boldsymbol{V = -L\frac{\Delta I}{\Delta t}} \tag{30}$$

$\boldsymbol{L}$ is called **self inductance**. Unit is $\mathbf{H}$ (**henry**).

$L = \frac{\mu N^2 S}{l}$

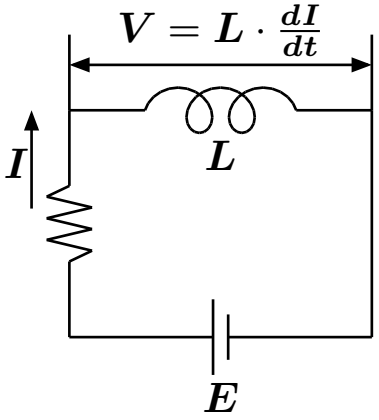
$= \mu n^2 Sl$

($n = \frac{N}{l}$ is number of roll per unit length of coil)

This voltage $\boldsymbol{V}$ acts as a potential slope that impedes changes in current ($\boldsymbol{\Delta I}$). This phenomenon is called **Self-induction**.

Thus, coil softens(impede) rapid current changes and the ability

4.1.b Energy of Coil



$W = IVt$ (V is self inductance)

The work during dt is $I \cdot L\frac{dI}{dt} \cdot dt$

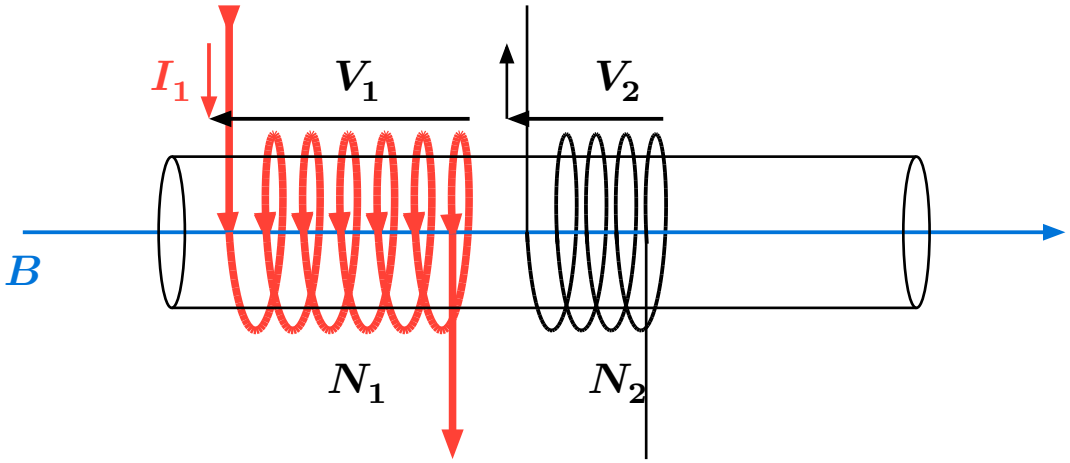
Thus, $\int_0^I ILdI = \frac{1}{2}LI^2$

Therefore, Energy U is

$$\boldsymbol{U = \frac{1}{2}LI^2} \tag{31}$$

(コイルにエネルギーってあんまり馴染ないけど, 超伝導でもないとすぐ消失するもんな. 逆に超伝導の違和感がすごい)

4.1.c Mutual Induction



$B = \mu\frac{N_1}{l} \cdot I_1 \quad \Leftrightarrow \quad \Delta\Phi = \mu\frac{N_1}{l}S \cdot \Delta I_1$

$V_2 = -N_2\mu\frac{N_1}{l}S \cdot \frac{\Delta I_1}{\Delta t}$

Let $N_2\mu\frac{N_1}{l}S = \boldsymbol{N_1 N_2} \cdot \frac{\mu}{l}\boldsymbol{S} = \boldsymbol{M}$,

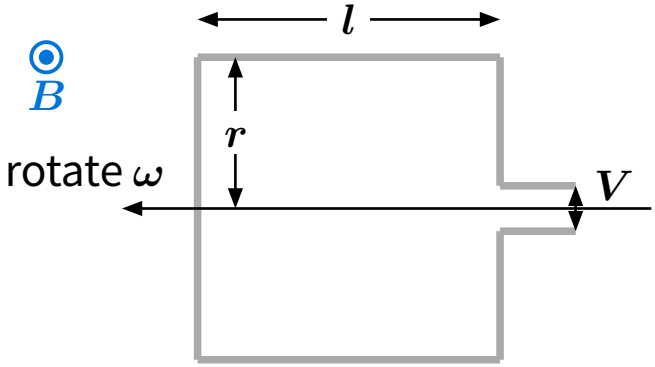
$$\boldsymbol{V_2 = -M\frac{\Delta I_1}{\Delta t}} \tag{32}$$

$\left(V_1 = -M\frac{\Delta I_2}{\Delta t}\right)$

M is called **mutual inductance.**

4.2 Alternating Current

4.2.a How to make A.C.



$V = 2vBl\sin\theta, v = r\omega$ Thus,

$$\boldsymbol{V = 2r\omega Bl \cdot \sin\omega t} \tag{33}$$

4.2.b Effective value

(かなり遅れてるので図はスキップ．ごめんなさいにょ)

$I = \frac{V}{R} = I_0 \sin \omega t)$ (I_0 is max current)

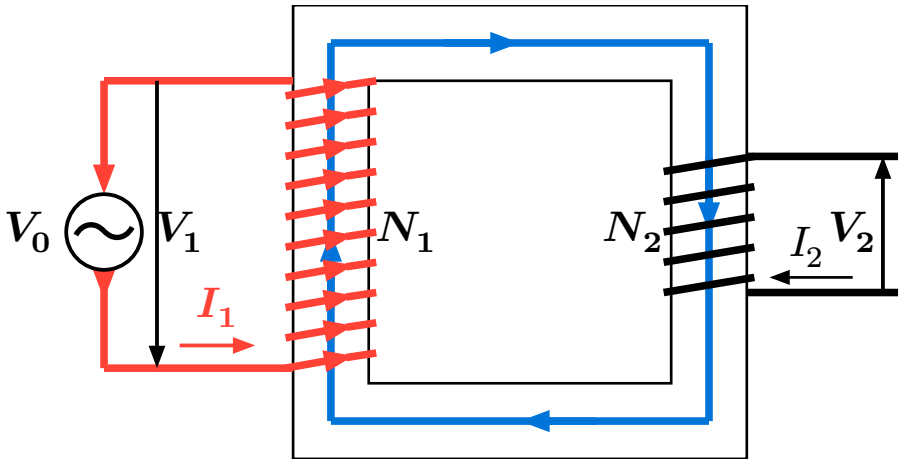
$P = IV = I_0 V_0 \sin^2 \omega t = \frac{I_0 V_0}{2}(1 - 2\cos \omega t)$

Therefore, mean power(effective power) $\bar{P} = \frac{1}{2} I_0 V_0$

effective voltage and current is

$$\begin{gathered} \boldsymbol{V_e = \frac{1}{\sqrt{2}} V_0}, \\ \boldsymbol{I_e = \frac{1}{\sqrt{2}} I_0}, \\ \left(\boldsymbol{\bar{P} = I_e V_e}\right) \\ \boldsymbol{V_e = R I_e} \end{gathered} \tag{34}$$

4.2.c Transformer



(鉄芯に沿って磁束がループを描くことが前提．$\mathbb{B}$は厳密に$\mathbb{H}$の scalar 倍ではなく，鉄の強い磁化によって vector が歪められる．ってことは Equation 27 のあたり嘘か…)

coil のまく向き注意！この note に書いた Faraday の法則は左回転(巻く過程での全体的な進行方向を回転ベクトルとした巻き方)でのもの．

$V_1 = -N_1 \cdot \frac{\Delta \Phi}{\Delta t}, \quad V_2 = -N_2 \cdot \frac{\Delta \Phi}{\Delta t}$

Therefore,

$$\boldsymbol{V_1 : V_2 = N_1 : N_2} \tag{35}$$

4.2.d A.C and Coil

When $I_L = I_{L0}\sin\omega t$,

$$\begin{aligned} V_L &= L\cdot\frac{dI_{L0}}{dt} \quad (\because \text{Equation 30}) \\ &= L\cdot I_{L0}\omega\cos\omega t \\ &= I_{L0}X_L\cos\omega t \end{aligned}$$

$$\boldsymbol{I_L = I_{L0}\sin\omega T \quad \Leftrightarrow \quad V_L = I_{L0}X_L\cos\omega t} \tag{36}$$

$X_L = \frac{V_{L0}}{I_{L0}} = \omega L$ is called **reactance**. the same unit $[\Omega]$ is used for this. **This is not resistance**, bacause no energy is consumed in A.C. current. (Joule heat が発生しない上に，coiil にたまる energy は平均的に 0 だから．reactance は電圧から電流への応答性であって，energy を逃がす resistance とは本質的に異なる)

4.2.e A.C and Capacitor

When$V_C = V_{C0}\sin\omega t$,

$$\begin{aligned} I_C &= \frac{dQ}{dt} \quad (\because \text{difinition of current}) \\ &= C\cdot\frac{dV_C}{dt} \\ &= C\omega V_{C0}\cos\omega t \end{aligned}$$

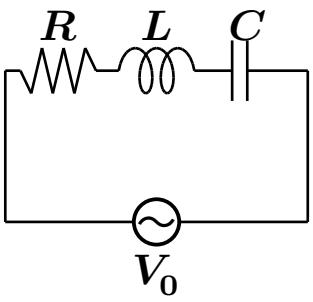
Here, Set $I_C = I_{C0}\sin\omega t$,

$$V_C = -I_{C0}X_C\cos\omega t$$

$$\boldsymbol{I_C = I_{C0}\sin\omega t \quad \Leftrightarrow \quad V_C = -I_{C0}X_C\cos\omega t} \tag{37}$$

reactance of capacitor $\boldsymbol{X_C = \frac{1}{\omega C}}$

4.2.f Impedance



Since Current I must not delay in the circuit,

Let $I = e^{i\omega t}$

$V_R = Re^{i\omega t}$,

$V_L = i\omega L e^{i\omega t}$,

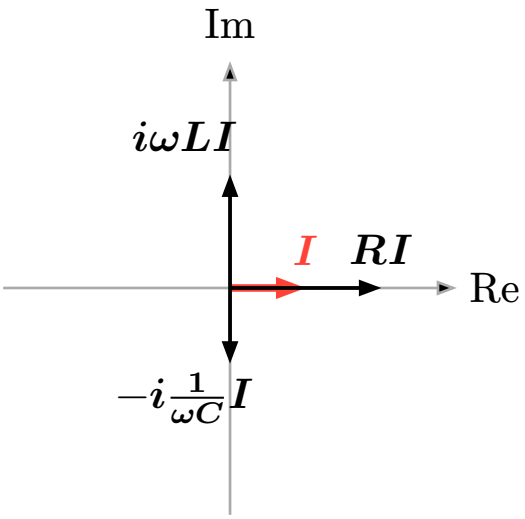
$V_C = -i\frac{1}{\omega C}e^{i\omega t}$

Thus,

$\boldsymbol{V_R + V_L + V_C = I \cdot \left(R + i\omega L - i\frac{1}{\omega C}\right)}$

$$\boldsymbol{V_0 = I_0 \cdot \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

$$\boldsymbol{= I_0 Z}$$



Let phase difference be φ,

$\boldsymbol{\tan\varphi = \frac{\omega L - \frac{1}{\omega C}}{R}}$

By the way, when the connection is parallel, $Z' = \frac{1}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}, \tan\varphi = -\frac{\omega L - \frac{1}{\omega C}}{R}$)

Since only resistor create Joule heat , power consumption will be like this:

$\bar{P} = RI_e^2$

Here $R = Z\cos\varphi$,

$$\boldsymbol{\bar{P} = I_e V_e \cos\varphi} \tag{38}$$

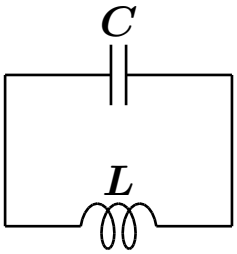
$\cos\varphi$ is called **power factor**.

4.2.g Resonance Circuit

In RLC-circuit(Figure 7). When $\omega L = \frac{1}{\omega C} \quad \Leftrightarrow \quad \boldsymbol{\omega = \frac{1}{\sqrt{LC}}}$,
$Z = R$ and impedance will be minimamized. Thus, Huge current flows at specific frequency. This frequency is called **resonance frequency**.

(capacitor の reactance と coil の reactance が等しいから，電圧に対する電流の応答の強さが双方で同じで位相差 πなため，全 energy をやりとりできる？電圧に対する位相差 0 でピッタリ合った電流変化が得られるのか)

4.2.h Oscillation



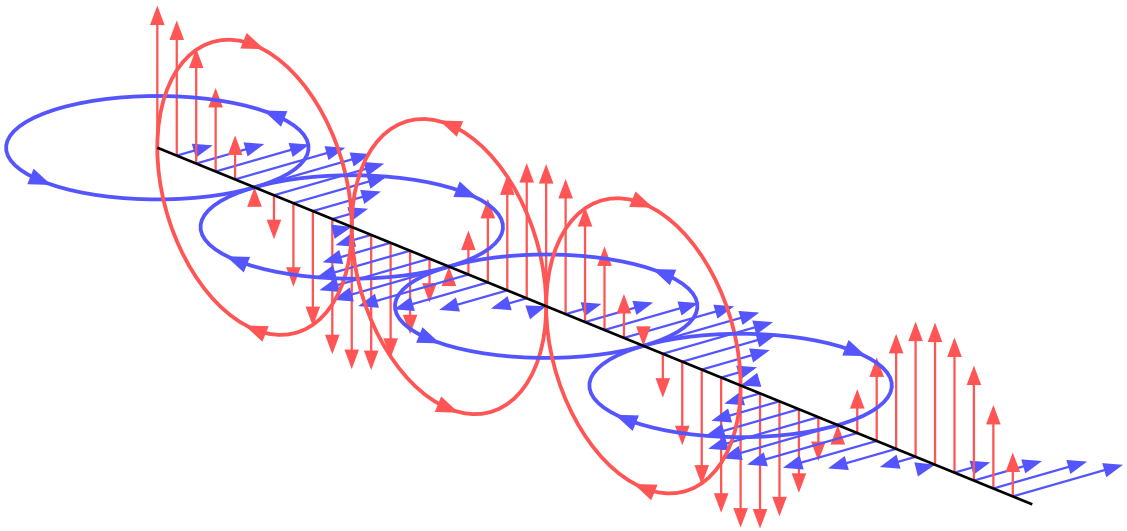
The current oscilate at $\omega = \frac{1}{\sqrt{LC}}$ in this circuit.

The energy is exchanged between coil and capacitor.

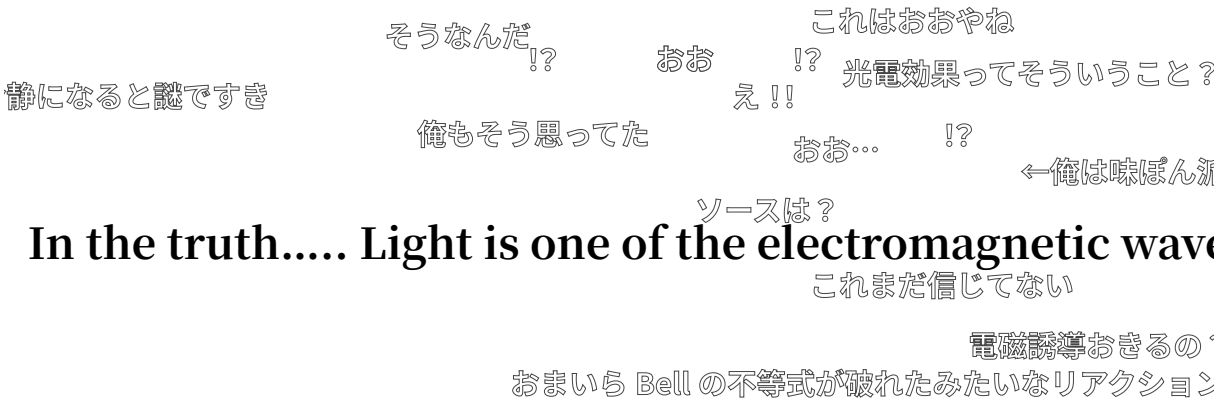
(上と同じことが起きているだけ)

4.3 Electromagnetic Waves

(あれ !! 電場ってなんだっけ@_@ ; となる．この Note の内容では電子の存在しない領域を電磁波が伝わる理由を説明できなさそう)

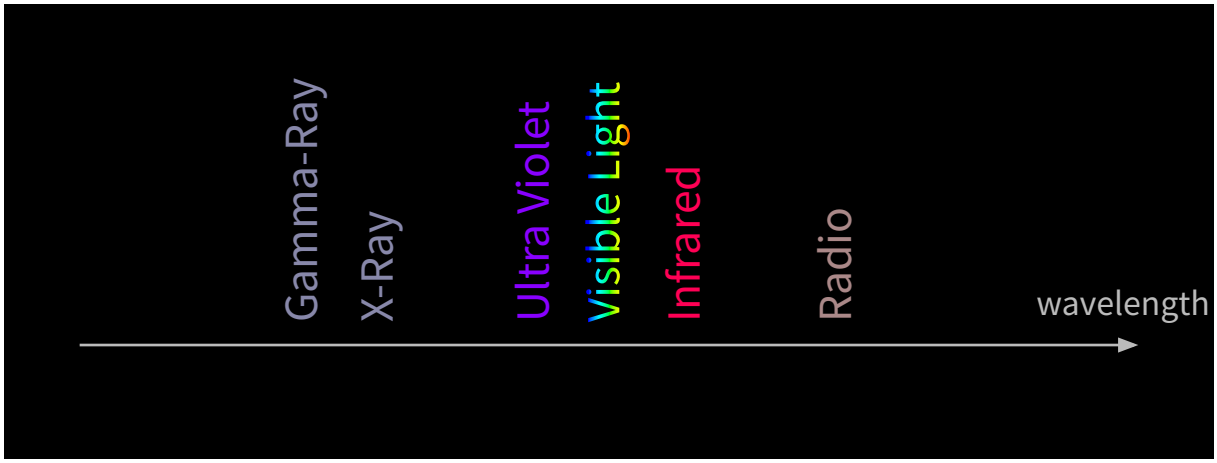


おおむねこんなイメージらしいです



(field の伝搬速度が光速なので，電磁波は光速で伝搬するのか)

(教員：分光は面倒だ，普通 amateur はどうやって表面温度を推定する) (ワイ：えあっ(動悸))



5 *Atom*

5.1 Electron

5.1.a Cathode-Rays

Gas Discharge – phenomenon of electrons flowing in gas. When pressure of gas is low, the entire gas emit light.

Cathode-rays = Electron Beam

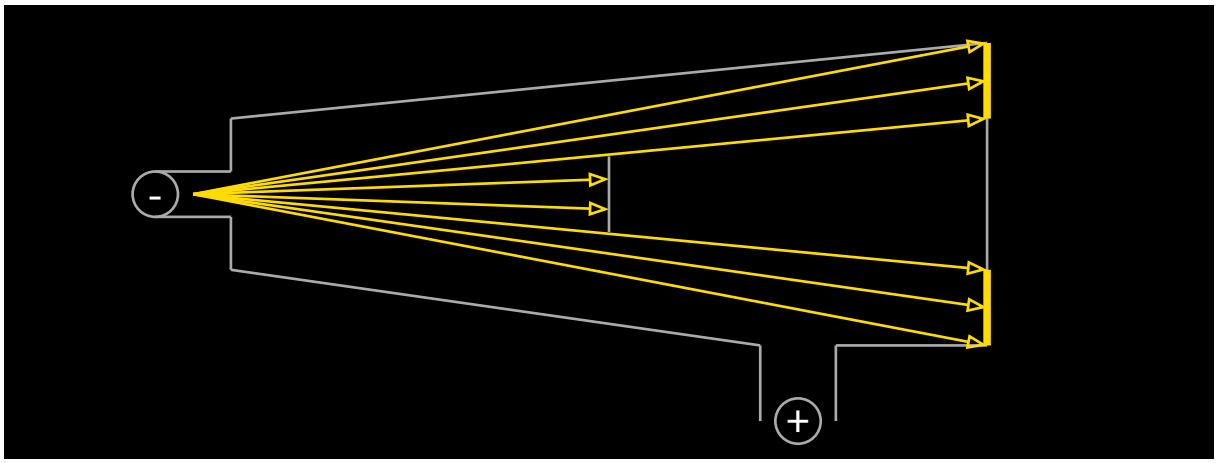


Figure 8: when a cathode-ray collides fluorescent material, it emit light.

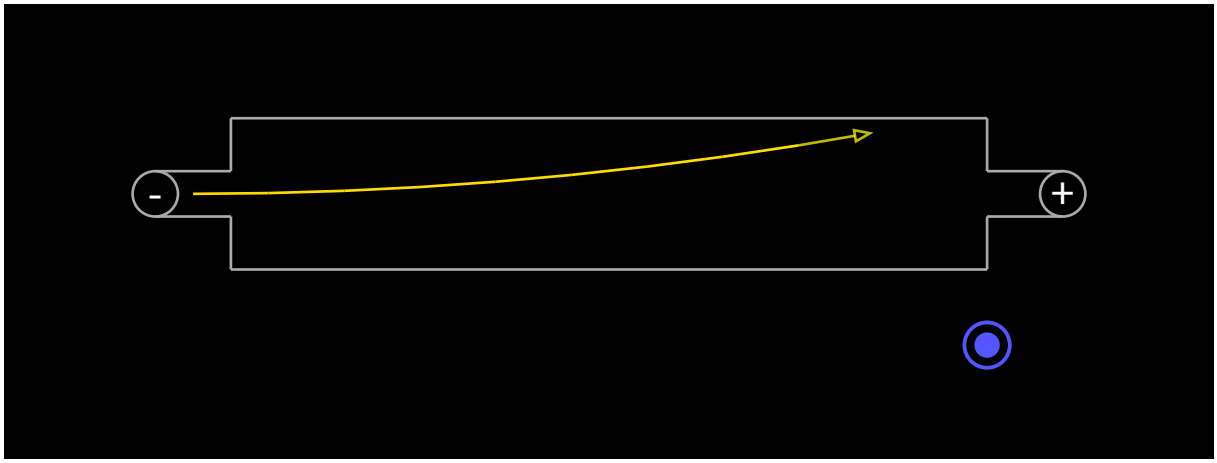
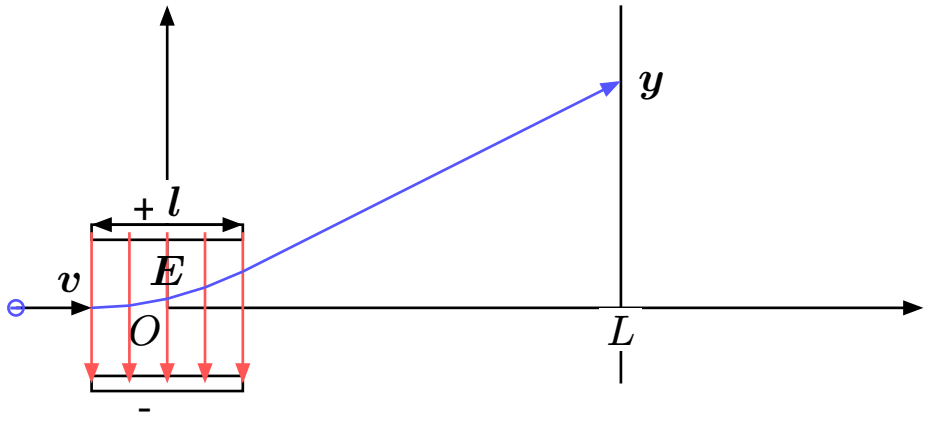


Figure 9: Cathode-rays receive the same force as lorentz force of negative-charge.

5.1.b How to measure the specific charge

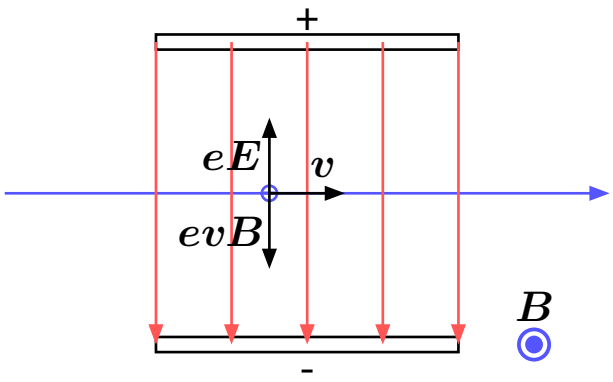
1.



$$y = \frac{1}{2}\frac{eE}{m}\frac{l^2}{v^2} + \left(L - \frac{l}{2}\right) \cdot \frac{eE}{m}\frac{l}{v^2} = \frac{eEl}{mv^2} \cdot \left(\frac{1}{2}l + L - \frac{1}{2}l\right)$$

$$\boldsymbol{y = \frac{e}{m}\frac{ElL}{v^2}}$$

2.



When $v = \text{const.}$,

$eE = evB \quad \Leftrightarrow \quad v = \frac{E}{B}$

We got v.

Therefoer $\frac{e}{m} = \frac{yE}{B^2lL}$

$\frac{e}{m}$ is called **specific charge.**

5.1.c Measuring the value of e

Method : Let m as measurable(large).

1. By irradiating X-ray at molecule of air, make ion.
2. attach the ion to particle of oil.
3. measure the mass of oil (utilize terminal velocity in air)
4. measure the charge of oil q (utilize electric field)
5. do 1 – 4 a lot.
6. find the minimum difference of q –> **elementary charge**

5.2 Particle-likeness of Light

5.2.a Photoelectric Effect

When a light is irradiated to metal, if it's frequency is over a threshold, electrons fly out from the metal.
(**not depending it's amplitude and irradiating time**)

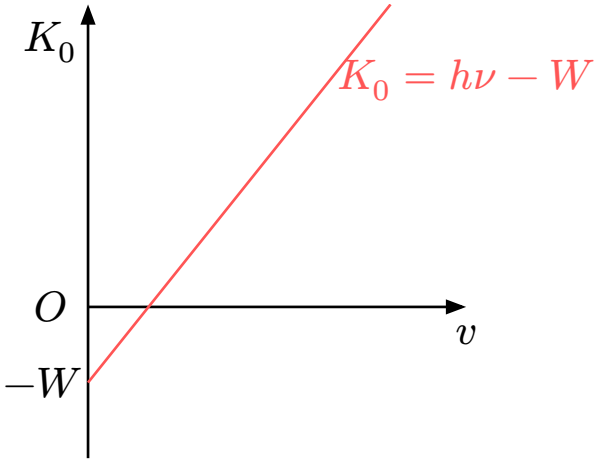
5.2.b Light quantum Hypothesis

$$\boldsymbol{E = h\nu} \tag{39}$$

$\boldsymbol{E}$ is Energy
$\boldsymbol{h}$ is **Planck Constant** $\approx 6.63 \cdot 10^{-34}\ \mathrm{J\ s}$
$\boldsymbol{\nu}$ is frequency of light.

If light consists of particles which has only parameter energy
(**photon**) and the energy depends on only it's frequency, we can explain the Photoelectric Effect.

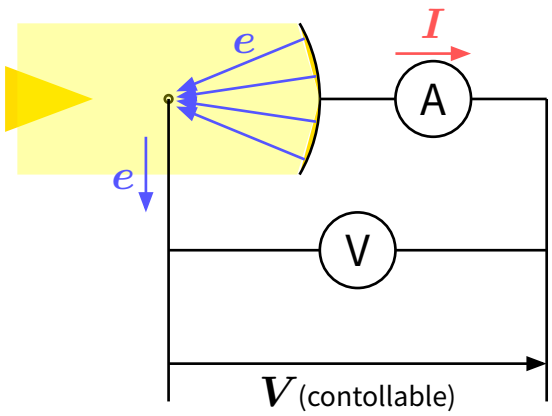
(if light is patricle , the probability of "Two photon colide to an electron almost simultaneously." is nealy zero. Thus, if energy of electron is lower than a threshold , no electrons fly out.)

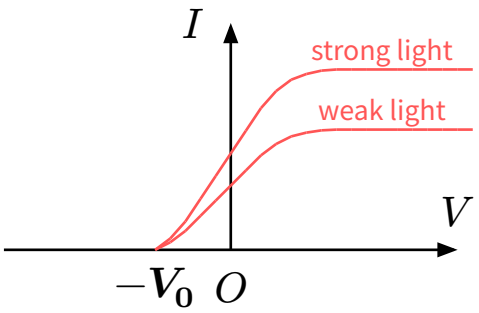


K_0 is maximum kinetic energy of an electron.

W is called **work function** which is minimum work to take an electron to infinity. (「金属結晶中の自由電子のうち最も出てこやすいやつ」の potential の-1 倍)

量子の粒子性は相互作用をするときにのみ現れる.





1. make an electron current by light
2. make a potential slope by battery

When voltage is $-V_0$, $I = 0$ i.e. kinetic energy of electrons is 0.

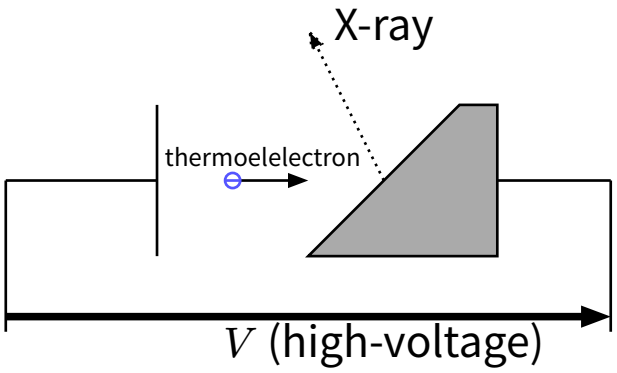
Set first kinetic energy of an electron to K_0,

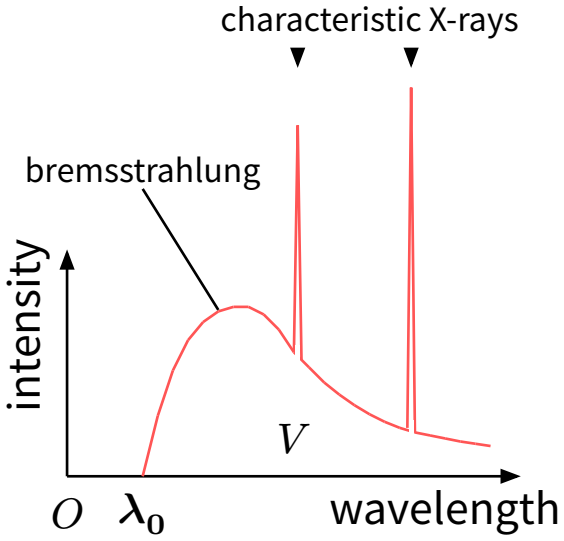
$\boldsymbol{K_0 = eV_0}$

5.2.c Electron Volt

1 eV equals to " **Kinetic energy of an electron gained by voltage V**" (eV ってそのままかよ)

5.3 X-Ray





$$\lambda_0 = \frac{hc}{eV} \tag{40}$$

(これが蛍光 X 線分析か)

5.3.a Bragg's Law

Irradiating X-ray at cristals from point source, we can obrain X-ray diffraction images. In these images, **Laue spot** can be seen.

Consider parallel planes in a crystal containing atoms arranged regularly, spaced by a distance d. Let θ be the angle that the X-rays make with the planes. The reflected light constructively interferes under the following condition:

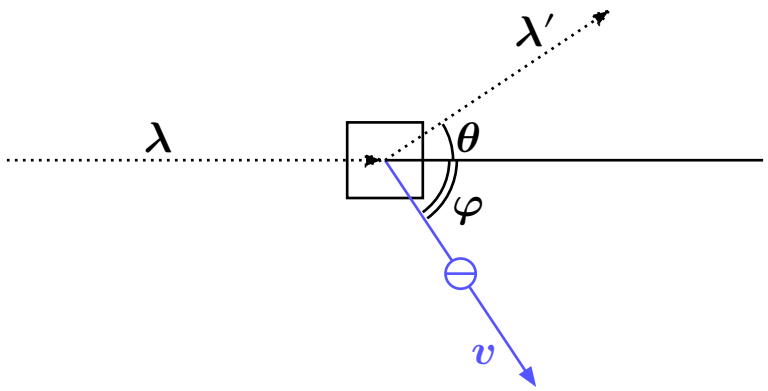
$$2d\sin\theta = n\lambda \tag{41}$$

This is known as **Bragg's condition**.

(反射前と後があるので$2d\sin\theta$) (X-ray diffraction image は 3 次元だから十分な情報量ってこと？すごいね)

5.3.b Compton Effect

When X-rays are irradiated into a substance, the wavelength of the scattered X-rays increases depending on θ.



This phenomenon can be explained if we assume X-ray to have **particle-likeness.**

Photon is supposed to have momentum

$$\boldsymbol{p = \frac{h\nu}{c}\left(= \frac{h}{\lambda}\right)} \tag{42}$$

これマジですか

If this is elastic collision, letting m be mass of an electron,

$$\begin{cases} \begin{pmatrix} \frac{h}{\lambda} \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{h}{\lambda'}\cos\theta + mv\cos\varphi \\ \frac{h}{\lambda'}\sin\theta - mv\sin\varphi \end{pmatrix} \\ \dfrac{hc}{\lambda} = \dfrac{hc}{\lambda'} + \dfrac{1}{2}mv^2 \end{cases}$$

$$\Rightarrow \begin{cases} m^2v^2(\cos^2\varphi + \sin^2\varphi) = \dfrac{h^2}{\lambda^2} - \dfrac{h^2\cos\theta}{\lambda\lambda'} + \dfrac{h^2}{\lambda'^2}(\cos^2\theta + \sin^2\theta) \\ v^2 = \dfrac{2hc}{m}\left(\dfrac{1}{\lambda} - \dfrac{1}{\lambda'}\right) \end{cases}$$

$$\Leftrightarrow 2mc\left(\tfrac{1}{\lambda} - \tfrac{1}{\lambda'}\right) = h\left(\tfrac{1}{\lambda^2} + \tfrac{1}{\lambda'^2} - \tfrac{2\cos\theta}{\lambda\lambda'}\right)$$

$$\Leftrightarrow 2mc\tfrac{\lambda'-\lambda}{\lambda\lambda'} = h\tfrac{\lambda^2+\lambda'^2+2\lambda\lambda'\cos\theta}{\lambda^2\lambda'^2}$$

$$\Leftrightarrow \tfrac{2mc}{h}(\lambda'-\lambda) = \tfrac{\lambda'}{\lambda} + \tfrac{\lambda}{\lambda'} + 2\cos\theta$$

$$\approx 2 + 2\cos\theta$$

$$\boldsymbol{\lambda' - \lambda \approx (1+\cos\theta)\cdot\frac{h}{mc}} \tag{43}$$

5.4 Wave-Likeness of Particle

Wave-Particle Duality

De Broglie's hypothesis : a particle which has mass m, velocity v has wave-likeness. This wave is called **matter wave**.

De Broglie wavelength:

$$\boldsymbol{\lambda = \frac{h}{p} = \frac{h}{mv}} \tag{44}$$

($\boldsymbol{p}$ is momentum)

wavelength of electron which has potential V is $\boldsymbol{\lambda = \frac{h}{\sqrt{2meV}}}$

wavelength can be changed by V !

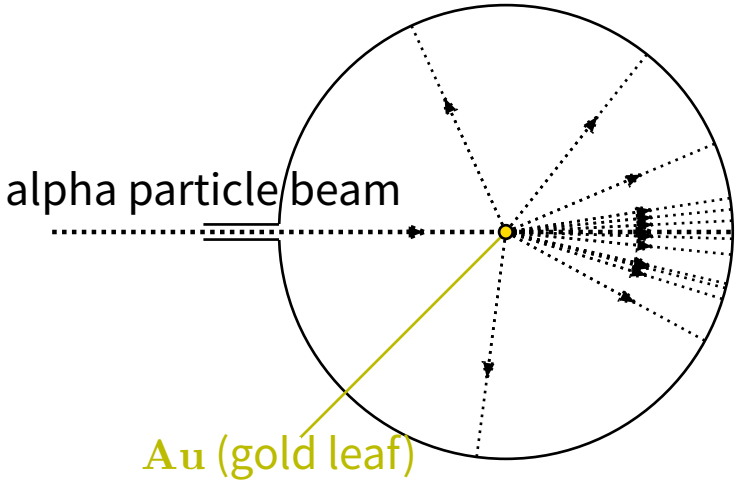
5.4.a Quantum Mechanics

Uncertainty Principle — Both position x and momentum $m\dot{x}$ of particle can not be decided accurately at the same time.

Position and momentum is trade off.

5.5 Atom and Nucleus

5.5.a Model of the Atom



The result implied that an atom have a small **nucleus** with a poisitive charge. Rutherford thought of **Rutherfords's model of the atom**. The model consists of a nucleus and electron revolving aroud the nucleus.

5.5.b Spectrum of Hydrogen Atom

Balmer found regularity of line spectrums of hydrogen. This is called **Balmer series**. Other set of line spectrums in different wavelength region, **Lyman series** and **Paschen series** was found.

$$\boldsymbol{\lambda^{-1} = R(n^{-2} - (n+1)^{-2})} \quad (n \in \mathbb{N}) \quad (R \approx 1.10 \cdot 10^7 \mathrm{m}^{-1}) \quad (45)$$

5.5.c Bohr's theory

Rutherford's model of the atom had a critical problem :

- radiating electromagnetic wave(bremsstrahlung), electron loses kinetic energy and slow down. This cause collision of electron and nucleus, showing consequence **Atom is Instable**. (This doesn't match the observation.)

- The model cannot explain characteristic X-rays (and other line spectrums)

(いまさらだけどもしかして電子って仕事 0 で曲げられないのか)

Bohr built a condition.

$$\boldsymbol{mvr = n \cdot \frac{h}{2\pi}} \quad (\boldsymbol{n} \in \mathbb{N}) \tag{46}$$

(m is mass of an electron. v is speed of it, r is radius of it's orbit.)

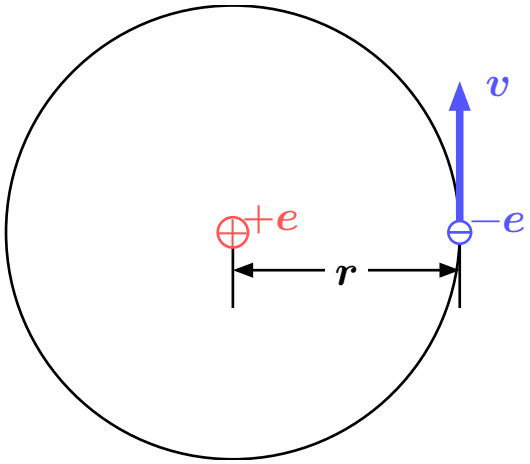
This is called **quantum condition.**

When this equation is satisfied, electron is state(steady state)

mvr : angular momentum of an electron

→ **angluar momentum of electron is discrete.**

5.5.d Energy State



When steady state,

$m\frac{v^2}{r} = k_0\frac{e^2}{r^2}$

$v^2 = k_0\frac{e^2}{rm} = \frac{n^2h^2}{4\pi^2m^2r^2})$ $\quad (\because \text{Equation } 46)$

$\frac{k_0e^2}{m} = \frac{n^2h^2}{4\pi^2m^2r}$

$$\boldsymbol{r = \frac{h^2}{4\pi^2k_0me^2} \cdot n} \quad (\boldsymbol{n} \in \mathbb{N}) \tag{47}$$

Consider about energy,

$$\boldsymbol{E} = \frac{1}{2}mv^2 - k_0\frac{e^2}{r} = \frac{1}{2}k_0\frac{e^2}{r} - k_0\frac{e^2}{r} = -\frac{\boldsymbol{k_0e^2}}{\boldsymbol{2r}}$$

$$= -\frac{2\pi^2k_0^2me^4}{n^2h^2} \quad (\because \text{Equation } 47)$$

$$\boldsymbol{E_n = -\frac{2\pi^2k_0^2me^4}{h^2}\cdot n^{-2}} \quad (\boldsymbol{n} \in \mathbb{N}) \tag{48}$$

$(n = 1)$ — **ground state**

$(n \geq 2)$ — **exiteed state**

Energy of electron is proportional to n^{-2}. Line spectrum of hydrogen atom can be explained with this.

$$\Delta E = h\frac{c}{\lambda} = \frac{2\pi^2k_0^2me^4}{h^2}\left(n^{-2} - (n+1)^{-2}\right)$$

$$\lambda^{-1} = \frac{2\pi^2k_0^2me^4}{h^3c}\left(n^{-2} - (n+1)^{-2}\right) = R\left(n^{-2} - (n+1)^{-2}\right)$$

(おお(line spectrum を説明するために quantum condition を導入した感があって感動はあまりない人))

20261106 の分は試験をうけるために授業を休んだため，教科書から補完

5.5.e Nucleus

nuclear force — strong force between nucleons(proton and neutron). This cluster nucleons together in narrow space(nucleus).

unified atomic mass unit — $\frac{1}{12}$ of mass of ${}^{12}_{6}\mathrm{C}$